

Seminar 4 - Exercises

- A firm has production function $y = 0.2 \ln(x_1) + 0.8 \ln(x_2)$.
 - Construct Lagrange function, derive first order conditions and derive relationship between amount of inputs ($MRTS$) and prices of inputs (w_1 and w_2).
 A: $F.O.C$: $w_1 = \lambda \frac{0.2}{x_1}$, $w_2 = \lambda \frac{0.8}{x_2}$, $y = 0.2 \ln(x_1) + 0.8 \ln(x_2)$, $TRS = -\frac{x_2}{4x_1} = -\frac{w_1}{w_2}$
 - Calculate conditional demand functions $x_1(w, y)$ and $x_2(w, y)$.
 A: $x_1 = e^y \left(\frac{w_2}{4w_1} \right)^{\frac{4}{5}}$, $x_2 = e^y \left(\frac{4w_1}{w_2} \right)^{\frac{1}{5}}$
- Consider Leontief production function in the form $y = \min\{ax_1, bx_2\}$
 - Prove that cost function equal $c(w, y) = y(\frac{w_1}{a} + \frac{w_2}{b})$ and derive conditional demand functions.
 A: Express conditional demands: $x_1 = \frac{y}{a}$, $x_2 = \frac{y}{b}$ and plug them into general form of cost function $c = w_1x_1 + w_2x_2$. *Note:* for perfect complements, the cost minimizing choice of inputs does not depend on the input prices.
 - What are minimal costs for x_1 and x_2 when $a = 3$, $b = 2$, $w_1 = 4$, $w_2 = 2$ and $y = 8$?
 A: $c = \frac{56}{3}$
- Earl sells lemonade in a competitive market on a busy street corner. His production function is $f(l, h) = l^{\frac{1}{3}}h^{\frac{1}{3}}$, where output is measured in liters. l is the number of kilos of lemons he uses, and h is the number of hours he spends squeezing them.
 - Does Earl have constant/decreasing/increasing returns to scale?
 A: decreasing
 - If he is going to produce y units of lemonade in the cheapest way possible, what is the number of kilos of lemons and the number of hours he will use?
 A: $l(w_l, w_h, y) = \sqrt{y^3 \frac{w_h}{w_l}}$; $h(w_l, w_h, y) = \sqrt{y^3 \frac{w_l}{w_h}}$
 - What is the cost to Earl of producing y units at factor prices w_l and w_h ?
 A: $c(w_l, w_h, y) = 2\sqrt{y^3 w_l w_h}$
- A university cafeteria produces square meals, using only one input and a rather remarkable production process. We are not allowed to say what that ingredient is, but an authoritative kitchen source says that "fungus is involved." The cafeteria production function is $f(x) = x^2$, where x is the amount of input and $f(x)$ is the number of square meals produced.
 - Does the cafeteria have constant/decreasing/increasing returns to scale?
 A: increasing
 - How many units of input does it take to produce 144 square meals? If the input costs w per unit, what does it cost to produce 144 square meals?
 A: 12; $c = 12w$
 - How many units of input does it take to produce y square meals?
 A: $\sqrt{y}$
 - If the input costs w per unit, what is the average cost of producing y square meals?
 A: $AC(w, y) = \frac{w}{\sqrt{y}}$
- For the production function $f(x_1, x_2) = (9x_1 + 18)^{\frac{1}{2}}(16x_2 + 32)^{\frac{1}{4}}$
 - Find demand functions $x_1(y, w_1, w_2)$ and $x_2(y, w_1, w_2)$ which minimize costs.
 A: $x_1 = \left(\frac{y}{6}\right)^{\frac{4}{3}} \left(\frac{2w_2}{w_1}\right)^{\frac{1}{3}} - 2$; $x_2 = \left(\frac{y}{6}\right)^{\frac{4}{3}} \left(\frac{w_1}{2w_2}\right)^{\frac{2}{3}} - 2$

- Find optimal demands x_1 and x_2 given required output level $y_0 = 18$ and input prices $w_1 = 6$ and $w_2 = 1$ which minimize costs.

A: $x_1 = 1, x_2 = 7$

- Express cost function $c(y, w_1, w_2)$.

A: $c(w_1, w_2, y) = \left(\frac{y}{6}\right)^{\frac{4}{3}} (w_1^2 w_2)^{\frac{1}{3}} \left(2^{\frac{1}{3}} + 2^{-\frac{2}{3}}\right) - 2(w_1 + w_2)$